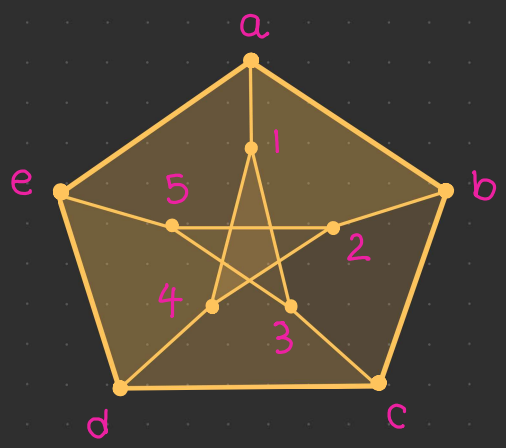
Q1

Show that the following two graphs are isomorphic by labelling the vertices of the right-hand graph appropriately and describing the associated bijective maps.

G



H



We begin by observing that both G and H contain 10 vertices and 15 edges. This is good news indeed, as our claimed isomorphism should preserve vertex adjacency. We also note that H is a representation of the Petersen graph, which is a fact that we will make use of when constructing a map between G and H. We identify a 5-cycle in G as the pentagon on vertices $\{a, b, c, d, e\}$, and relabel the vertices as follows:

$$A_1 = a \;,\; A_2 = b \;,\; A_3 = c \;,\; A_4 = d \;,\; A_5 = e \;.$$

Each of these 5-cycle vertices has a unique adjacent vertex that is not part of the 5-cycle,

namely those vertices in $\{1,2,3,4,5\}$, which we relabel as:

$$B_1 = 1,\ B_2 = 2,\ B_3 = 3,\ B_4 = 4,\ B_5 = 5,$$

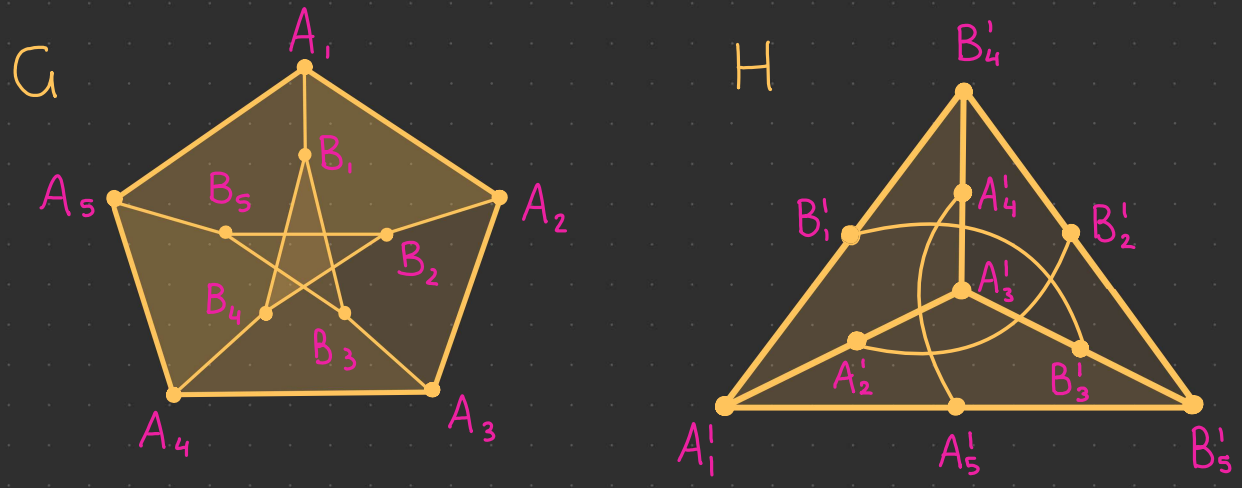
where each B_i is the inner vertex connected to A_i. We construct the bijective map

$$\Psi : G \to H$$

given by

$$\Psi(A_i) = A_i', \text{ and}$$
$$\Psi(B_i) = B_i' \quad \text{for} \quad 1 \leq i \leq 5$$

which can be represented as

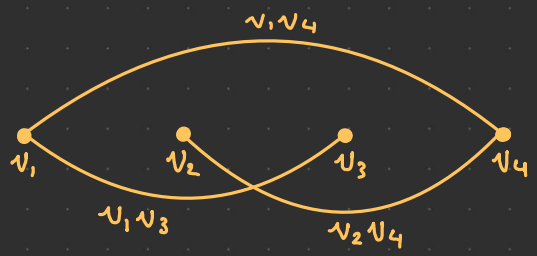


By inspection, we observe that the one-to-one correspondence between vertices in G and H induced by Ψ preserves adjacency relationships and hence we conclude, is an isomorphism.

Find a simple graph with at least one edge which is isomorphic to its complement. Is yours the smallest such graph (in terms of the number of vertices)? Why / why not?

Consider the path graph P_4, the simple graph on four vertices $\{v_1, v_2, v_3, v_4\}$, with edges $(v_1, v_2), (v_2, v_3), (v_3, v_4)$. The complement of P_4, which we will denote $\overline{P_4}$, is the graph on the same vertex set where two vertices are adjacent if and only if they are not adjacent in P_4. Thus, the edges of $\overline{P_4}$ are (v_1, v_3), (v_1, v_4), and (v_2, v_4) as shown below.





We define a bijective mapping $f: V(P_4) \to V(\overline{P_4})$, (where V denotes the set of vertices in P_4 and $\overline{P_4}$ respectively), by setting

$f(v_1) = v_1$,

$f(v_2) = v_3$,

$f(v_3) = v_2$, and

$f(v_4) = v_4$. This mapping preserves adjacency, since $(v_1, v_2) \in P_4$ corresponds to $(f(v_1), f(v_2)) = (v_1, v_3) \in \overline{P_4}$, and similarly for all other edges. Since f is a bijection and preserves adjacency, it is an isomorphism from P_4 to $\overline{P_4}$. Hence P_4 is self-complementary.

To address whether this is the smallest such self-complementary graph, consider a simple graph G on n vertices. Then the total number of possible edges in the complete graph K_n is $\binom{n}{2} = \frac{n(n-1)}{2}$. Since G and its complement $\overline{G}$ partition these edges, the sum of their edge counts must be

$$|E(G)| + |E(\overline{G})| = \binom{n}{2},$$

but since $G \cong \overline{G}$, we have that $|E(G)| = |E(\overline{G})|$. Letting $e = |E(G)| = |E(\overline{G})|$, we observe the relationship

$$\begin{aligned} e + e &= \binom{n}{2}, \\ \Rightarrow \quad 2e &= \binom{n}{2}, \\ \Rightarrow \quad e &= \frac{1}{2}\binom{n}{2}. \end{aligned}$$

Thus, we conclude that

$$|E(G)| = \frac{1}{2}\binom{n}{2},$$

which shows that any self-complementary graph must have exactly half of the possible edges among the n vertices. Since the number of edges must be an integer, this requires that $\binom{n}{2}$ is even, this leads to the necessary condition that $n \equiv 0 \bmod 4$ or $n \equiv 1 \bmod 4$. It follows that the smallest possible self-complementary graph with at least one edge has $n = 4$ vertices. Since P_4 is a self-complementary graph on four vertices, it is the smallest such construction.

Let G be a simple connected graph consisting of a cycle on n vertices. How many spanning trees does G have? How many non-isomorphic spanning trees does G have?

Consider a simple connected graph G that consists of a cycle on n vertices. That is, G is the cycle graph C_n, which consists of n vertices connected in a closed cycle, with each vertex having degree of exactly two.

To determine the number of spanning trees of G, we first recall that a spanning tree of G is a subgraph that is a tree and includes all n vertices of G. Since C_n consists of exactly n edges forming a single cycle, any spanning tree of G can be obtained by removing exactly one edge from this cycle. Since there are n edges in C_n, we obtain n distinct spanning trees by removing each edge in turn. Hence, C_n has exactly n spanning trees.

We now consider the number of non-isomorphic spanning trees of G. Each spanning tree of C_n consists of a single path on n vertices, obtained by deleting one edge from the cycle. Any such path is isomorphic to a path graph on n vertices P_n, which is structurally unique up to a relabelling of vertices. Therefore, all spanning trees of C_n are

structurally identical subgraphs. This implies that there is exactly one non-isomorphic spanning tree of C_n, namely the path graph P_n.

Thus, C_n has precisely n spanning trees, all of which belong to the same isomorphism class, defining exactly one non-isomorphic spanning tree.

Q2 Let T be a tree on k vertices. Prove, using induction, that if G is a simple, connected graph where every vertex has degree at least $k-1$, then T is isomorphic to some subgraph of G.

We proceed by induction on the number of vertices in the tree T.

For the base case, let $k = 1$. In this case, T consists of a single vertex with no edges. Since G is connected, it has at least one vertex. Consequently, we can identify a single vertex of G with the single vertex of T, and thus T is trivially isomorphic to a subgraph of G. This establishes the base case.

Assume for all integers $m \geq 1$ that every simple connected graph G with vertex degrees not less than $m-1$ contains a subgraph isomorphic to a tree on m vertices. We seek to show that this property holds for trees on $m+1$ vertices.

Let T be a tree on $m+1$ vertices. Since every tree with at least two vertices contains at least one leaf (vertex of degree 1), we choose a leaf v and let u be its unique adjacent vertex. Removing v from T results in a subtree T' on m vertices.

Now, let G be a simple connected graph in which every vertex has degree at least m. By our assumption, there exists a subgraph H of G that is isomorphic to T'. Let f be an isomorphic mapping from T' to H, and let $u' = f(u)$ be the image of u under f in H. Since u' is a vertex of G, and G has minimum degree at least m, it follows that u' has at least m neighbours in G. However, within H, the maximum number of neighbours u' can have is $m-1$ (since T has m vertices and is a tree). Therefore, there must exist a vertex $v' \in G$ that is adjacent to u', but does not belong to H.

We now extend the isomorphism f to include v by defining $f(v) = v'$. Since v is connected only to u in T and v' is adjacent to $u' = f(u)$ in G, this extended mapping preserves adjacency, thereby establishing an isomorphism between T and a subgraph of G.

By the principle of mathematical induction, we conclude that for all integers $K \geq 1$, any simple connected graph with minimum degree of at least $K-1$ contains as a subgraph every tree on K vertices.

Q3 Let f be a flow in a network N, and let $(U, \bar{U})$ be a cut in N. Let $A(U, \bar{U})$ denote the set of edges directed from vertices in U to vertices in $\bar{U}$. Prove that f is a maximum flow and $(U, \bar{U})$ is a minimum cut if and only if

$f(e) = c(e)$ for every edge $e \in A(U, \bar{U})$, and
$f(e) = 0$ for every edge $e \in A(\bar{U}, U)$.

We begin by recalling that a flow f in a network N is a function assigning a natural number value to each edge, subject to capacity constraints and flow conservation at intermediate vertices. A cut $(U, \bar{U})$ is a partition of the vertex set of N such that the source s belongs to U and the sink t belongs to $\bar{U}$. The capacity of the cut is defined as

$$c(U, \bar{U}) = \sum_{e \in A(U, \bar{U})} c(e).$$

where $A(U, \bar{U})$ denotes the set of edges directed from vertices in U to vertices in $\bar{U}$

By lemma 1.1, the value of the flow is given by

$$val(f) = f(U, \bar{U}) - f(\bar{U}, U),$$

where $f(U, \bar{U})$ is the total flow assigned to edges directed from vertices in U to vertices in $\bar{U}$, and similarly for $f(\bar{U}, U)$.

Now, suppose that f is a maximum flow and $(U, \bar{U})$ is a minimum cut. By theorem 1.4 (The max-flow Min-cut Theorem), the value of the maximum flow equals the capacity of the minimum cut such that

$$val(f) = c(U, \bar{U}).$$

Since $val(f) = f(U, \bar{U}) - f(\bar{U}, U)$, we find

$$f(U, \bar{U}) - f(\bar{U}, U) = c(U, \bar{U}).$$

By corollary 1.2 we have that $f(U, \bar{U}) \leq c(U, \bar{U})$, and since all flow values must be non negative, the only way for this equation to hold is if

$$f(U, \bar{U}) = c(U, \bar{U}), \text{ and} \quad (\star)$$

$$f(\bar{U}, U) = 0. \quad (\ddot\smile)$$

For $(\star)$ to hold, for every edge $e \in A(U, \bar{U})$ we must have $f(e) = f(c)$ since $f(e) \leq f(c)$ and here the total flow values on edges directed from vertices in U to vertices in $\bar{U}$ must reach the total capacity. The second condition, $(\ddot\smile)$, implies that for every edge $e \in A(\bar{U}, U)$, we must have $f(e) = 0$, meaning that there is zero flow being directed from vertices in $\bar{U}$ to vertices in U.

Conversely, suppose that $f(e) = c(e)$ for all $e \in A(U, \bar{U})$

and $f(e) = 0$ for all $e \in A(\bar{U}, U)$. Then the total flow value on vertices directed from U to $\bar{U}$ equals the total capacity such that

$$f(U, \bar{U}) = \sum_{e \in A(U, \bar{U})} f(e) = \sum_{e \in A(\bar{U}, U)} c(e) = c(U, \bar{U}).$$

Similarly, because all flow values from vertices in $\bar{U}$ to vertices in U are zero, we have that

$$f(\bar{U}, U) = 0$$

$$\Rightarrow \quad \text{val}(f) = f(U, \bar{U}) - f(\bar{U}, U) \quad \text{(by lemma 1.1)}$$

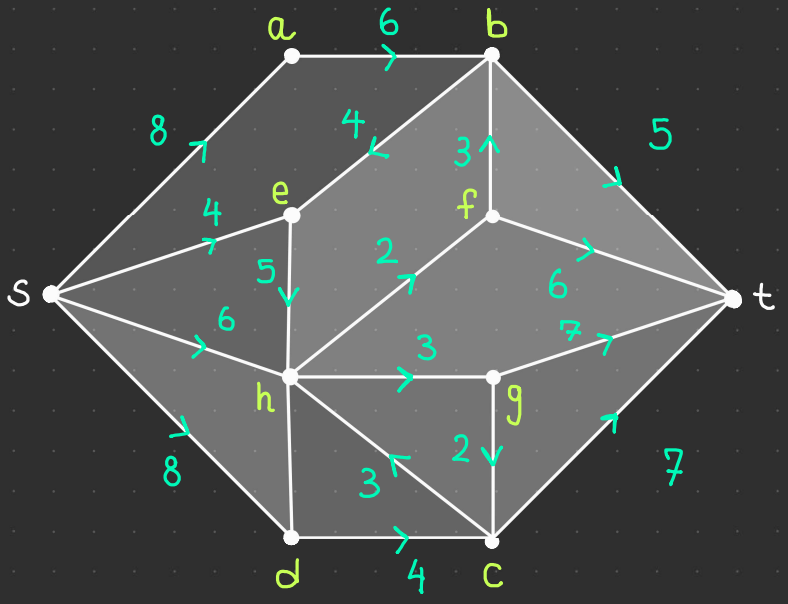
$$= c(U, \bar{U}) - 0$$

$$= c(U, \bar{U}).$$

Since the max flow value equals the capacity of the cut $(U, \bar{U})$, Theorem 1.4 tells us that the cut is a minimum cut and the flow is a maximum flow.

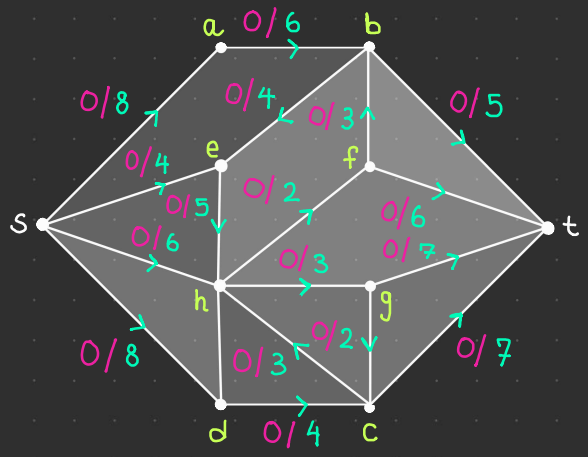
Thus, f is a maximum flow and $(U, \bar{U})$ is a minimum cut if and only if $f(e) = c(e)$ for every edge directed from vertices in U to vertices in $\bar{U}$, and $f(e) = 0$ for flow values directed from vertices in $\bar{U}$ to vertices in U.

Q4 Consider the following network N where the edge weights are the edge capacities.



(i) Use the Ford-Fulkerson Algorithm to find a maximum flow from s to t.

(1) Initialise network flow to zero:



(2) While flow incrementing paths exist from s to t, increment flow accordingly

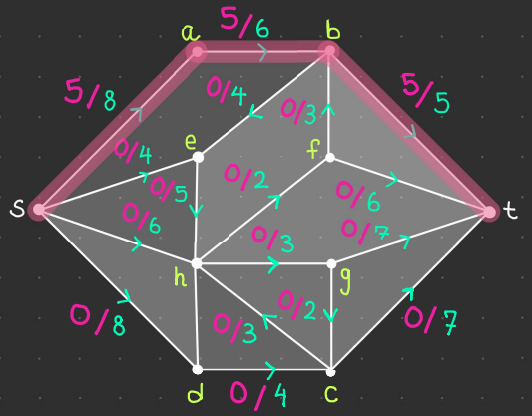
Iteration 1

Flow incrementing path $p = sabt$

Increment $i(P) = 5$

Value of flow $= val(f) + i(P)$

$= 0 + 5 = 5$



Iteration 2

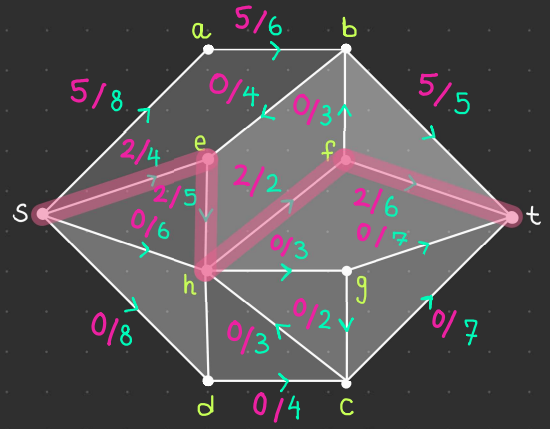
Flow incrementing path $p = sehft$

Increment $i(P) = 2$

Value of flow $= val(f) + i(P)$

$= 5 + 2$

$= 7$



Iteration 3

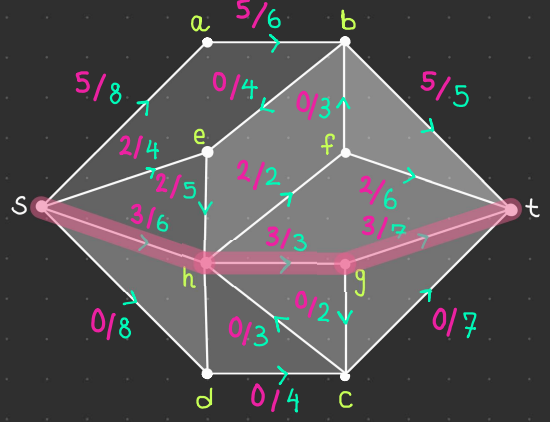
Flow incrementing path $p = sngt$

Increment $i(P) = 3$

Value of flow $= val(f) + i(P)$

$= 7 + 3$

$= 10$



Iteration 4

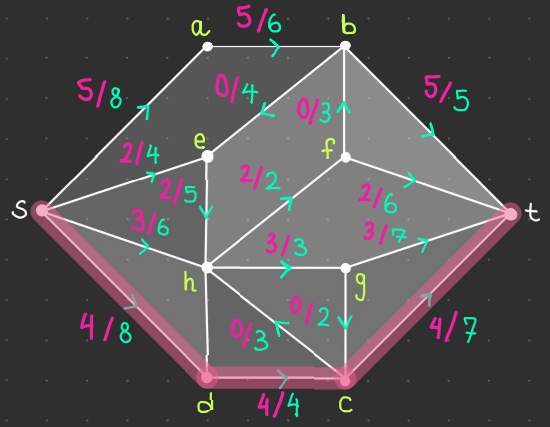
Flow incrementing path $p = sdct$

Increment $i(P) = 4$

Value of flow $= val(f) + i(P)$

$= 10 + 4$

$= 14$



No incrementing paths exist from s to t after iteration 4, and so we claim that the maximum flow on the network is 14.

(ii) Prove that this a maximum flow by using the Max-Flow Min-Cut Theorem.

In (i) we claimed that 14 is the maximum flow on N. The minimum cut of N from (i)

is $\{s, a, b, e, h, d\}$ which has a capacity of 14. By the max-flow min-cut theorem, we have that the maximum value of a flow on a network is equal to the capacity of the minimum cut, and so we indeed find that 14 is the maximum flow on our network N.

Q5 Let $N = (V, E)$ be a flow network with source s and sink t, and let f be a maximum flow in N with the value

$$F = val(f).$$

Now, define a modified network N' by increasing the capacity of every edge $e \in E$ by an integer constant $\delta > 0$; that is for every edge e we have

$$c'(e) = c(e) + \delta.$$

(i) Prove that the new maximum flow value F' in N' satisfies

$$F + \delta \leq F' \leq F + \delta|E|.$$

We begin by considering how increasing every edge capacity by a uniform amount δ affects the maximum flow in the network. Since every edge's capacity increases by δ, the maximum capacity of any cut in N' increases at least by δ times the number of edges connecting vertices across each partition of the cut.

First, we establish the lower bound on F', such that $F' \geq F + \delta$. Consider a minimum s-t cut (S, T) in N with capacity equal to F, so we have

$$c(S, T) = T.$$

The total newly added capacity in this cut is given by the number of edges connecting vertices in S to vertices in T multiplied by δ, which is at least δ because the cut contains at least one edge in any nontrivial network. Thus, the capacity of the same cut in N' satisfies

$$c'(S,T) = c(S,T) + \delta|E(S,T)| \geq F + \delta,$$

so we have that $c'(S,T) = F' \geq F + \delta$, a lower bound on F'.

We now proceed to establish an upper bound on F' to support the claim that $F' \leq F + \delta|E|$. Since every edge has its capacity increased by exactly δ, the total new capacity of the network is at most the original capacity plus an increase of δ per edge. In particular, consider a minimum s-t cut of N', say (S', T'). The capacity of this cut in N' is at most the original capacity plus δ multiplied by the number of edges (as all edges contribute precisely δ more than in N). This gives

$$F' = c'(S',T') \leq c(S',T') + \delta|E| \leq F + \delta|E|.$$

Thus, we obtain the upper bound.
Together, these bounds establish that

$$F + \delta \leq F' \leq F + \delta|E|,$$

as required. Hence we see that increasing all edges by a uniform positive value ensures that the new maximum flow is bounded both above and below by expressions depending on that value and the number of edges in the network.

(ii) Let $(U, \bar{U})$ be a minimum cut in N and recall that $A(U, \bar{U})$ denotes the set of edges directed from vertices in U to vertices in $\bar{U}$. Let $|(U, \bar{U})| = |A(U, \bar{U})| = k$. Prove that if $(U, \bar{U})$ is also a minimum cut in N', then the maximum flow in N' is

$$F' = F + \delta k.$$

Provide necessary and sufficient conditions for this to occur.

Consider a minimum cut $(U, \bar{U})$ in N, and recall that this implies that the total capacity of edges in $A(U, \bar{U})$ equals the value of the maximum flow in N, that is,

$$c(U, \bar{U}) = F.$$

Since each edge $e \in A(U, \bar{U})$ has its capacity increased by δ, the total capacity of this cut in N' is

$$c'(U, \bar{U}) = c(U, \bar{U}) + \delta k = F + \delta k.$$

Now assume that $(U, \bar{U})$ remains a minimum cut in N'. By theorem 1.4 (Max-flow Min-cut), the value of the maximum flow in N' must equal the capacity of this cut, so

$$F' = c'(U, \bar{U}) = F + \delta K.$$

This confirms the result when the assumed condition on the minimum cut is satisfied.

To determine necessary and sufficient conditions for $(U, \bar{U})$ to remain a minimum cut in N', we must ensure that no other cut has a smaller capacity than $c'(U, \bar{U})$. If $(U, \bar{U})$ was a unique minimum cut in N, then after increasing all edge capacities uniformly, we expect it to remain the minimum cut unless some other cut becomes equal in capacity. More specifically, if another cut $(X, \bar{X})$ in N had a capacity only slightly larger than F, but contained fewer than K crossing edges, then after increasing all capacities by δ, its new capacity $c'(X, \bar{X}) = c(X, \bar{X}) + \delta |A(X, \bar{X})|$ could be smaller than $c'(U, \bar{U}) = F + \delta K$, contradicting the assumption that $(U, \bar{U})$ remains a minimum cut.

Thus, a sufficient condition for $(U, \bar{U})$ to remain a minimum cut in N' is that for all other cuts $(X, \bar{X})$,

$$c(X,\bar{X}) \geq c(U,\bar{U}),$$

with strict inequality whenever $|A(X,\bar{X})| < k$. A necessary condition is that no alternative minimum cut $(X,\bar{X})$ in N, with $c(X,\bar{X}) = F$, has strictly fewer edges contained in $A(X,\bar{X})$ than $A(U,\bar{U})$, since otherwise the modified capacity of $(X,\bar{X})$, given by $c'(X,\bar{X}) = F + \delta|A(X,\bar{X})|$ might fall below $F + \delta k$.

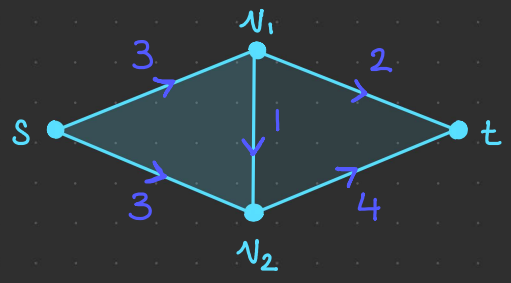
Therefore, $(U,\bar{U})$ remains a minimum cut in N' if and only if no other cut with a strictly smaller number of crossing edges was also a minimum cut in N. Under this condition, the maximum flow in N' satisfies $F' = F + \delta k$ as required.

(iii) Explain how uniformly increasing all edge capacities in N by δ might change which cut is minimum in N' when compared to N. Provide an example of a network where the minimum cut in N is not also a minimum cut in N'.

The effect of uniformly increasing all edge capacities in a network by δ can alter which cut is minimum because the relative differences between cut capacities may change. Since edge gains an additional capacity of δ, a cut that origionally

had relatively fewer crossing edges will see a proportionally smaller increase in capacity than a cut with more crossing edges. Consider the following network N, where the weights are edge capacities.

N



Let N contain the cuts $U = \{s, v_1\}$, $\bar{U} = \{v_2, t\}$, and $X = \{s, v_1, v_2\}$, $\bar{X} = \{t\}$.

Then $$c(U, \bar{U}) = c(sv_2) + c(v_1 t) + c(v_1 v_2)$$
$$= 3 + 2 + 1 = 6$$

and $$c(X, \bar{X}) = c(v_1 t) + c(v_2 t)$$
$$= 2 + 4 = 6.$$

Now, suppose we increase the capacity of each edge by δ in a modified network N'. Then the capacities of the cuts in N' become

$$c'(U, \bar{U}) = c'(sv_2) + c'(v_1 t) + c'(v_1 v_2)$$
$$= (3 + \delta) + (2 + \delta) + (1 + \delta)$$
$$= 6 + 3\delta$$

and $$c'(X, \bar{X}) = c'(v_1 t) + c'(v_2 t)$$
$$= (2 + \delta) + (4 + \delta)$$
$$= 6 + 2\delta.$$

While these cuts had equal capacities in N, we find that the cut $(X, \bar{X})$ in N' has a smaller capacity than the cut $(U, \bar{U})$ in N', meaning that $(U, \bar{U})$ is now the only minimal cut in N'. This example shows that increasing all edge capacities in a cut by the same amount can shift the balance among competing minimal cuts. We see that a previously minimal cut may cease to continue being minimal in the modified network if another cut with fewer crossing edges was close to it in capacity within the original network.